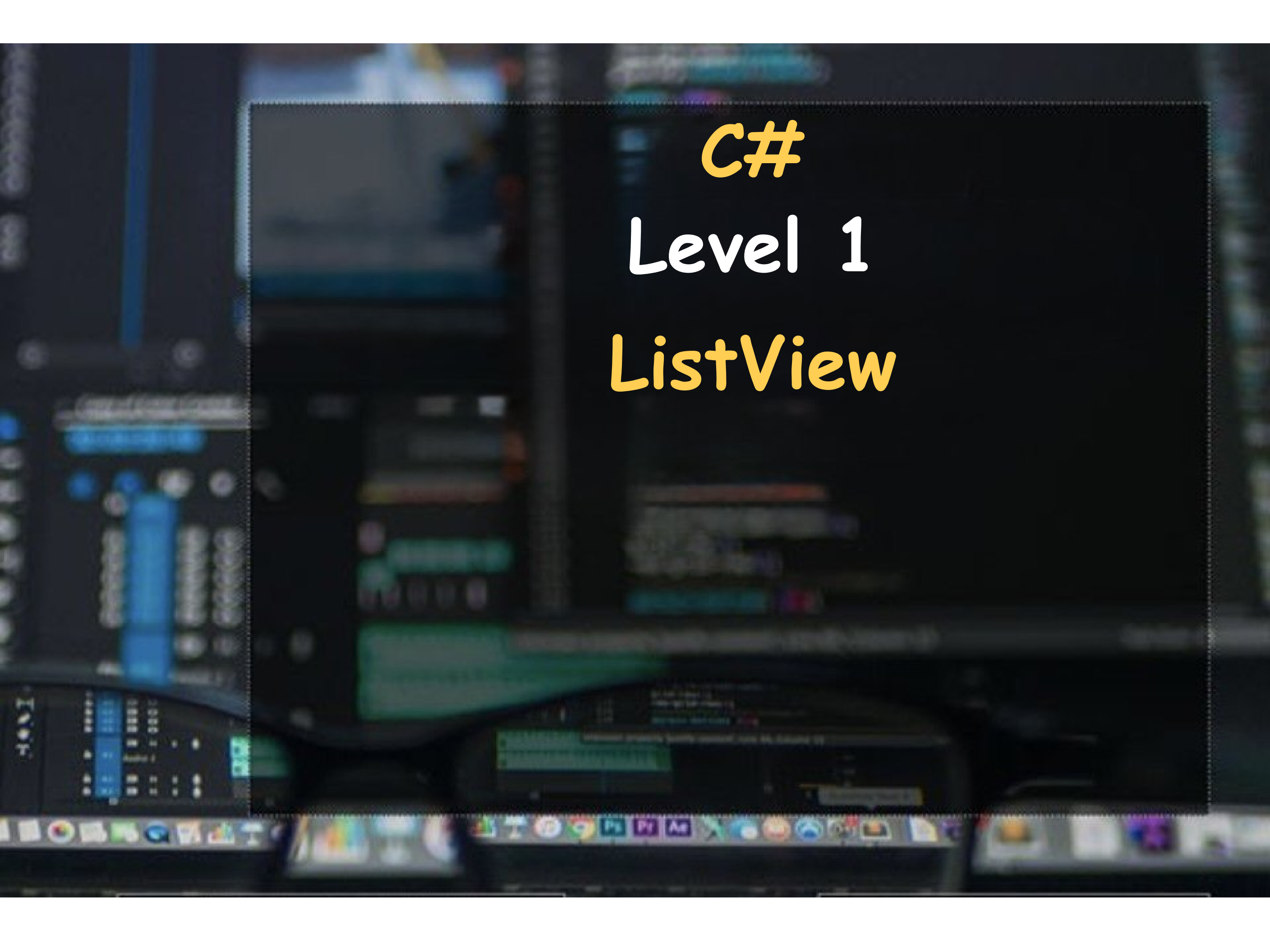




C#

Level 1

List View



C#

Level 1

ListView

Definition

ListView Control

 Windows Forms control used to display a collection of items in different layouts.

 In this project, it displays a person's ID, Name, and Gender image.

Explanation

Input Validation

-  Checks that ID and Name are not empty.
-  Checks that one Gender RadioButton is selected.

Duplicate ID Check

-  Loops through existing ListView items.
-  Compares the entered ID with existing IDs before adding a new person.

Creating the Item

-  ListViewItem is created using the ID.
-  ImageIndex selects the Gender image.
-  SubItems.Add() adds the Name.
-  Items.Add() adds the item to the ListView.

Procedure

1. Check ID and Name.
2. Check Gender selection.
3. Check whether the ID already exists.
4. Create a ListViewItem.
5. Set the Gender image using ImageIndex.
6. Add the Name using SubItems.Add().
7. Add the item using Items.Add().
8. Clear the inputs and reset Gender.
9. Focus on the ID TextBox.

💡 Example

```
ListViewItem item = new ListViewItem(textBox1.Text.Trim());  
item.ImageIndex = 1;  
item.SubItems.Add(textBox2.Text.Trim());  
listView1.Items.Add(item);
```

➡ Result

The person is added to the ListView with:

ID

Name

Gender image

Note

- ◆ ListViewItem represents one item/row.
- ◆ SubItems stores additional values for the same item.
- ◆ ImageIndex selects an image from the ImageList.
- ◆ Trim() removes spaces from the beginning and end of the input.



Comparison

ListViewItem

- ✓ Represents one item in the ListView.
- ✓ Contains the main value and additional SubItems.

SubItems

- ✓ Stores additional values for the same item.
- ✓ Commonly used for additional columns such as Name.

⊘ Warning

- ◆ Before using `listView1.SelectedItems[0]`, make sure an item is selected.
- ◆ Otherwise, accessing index 0 can cause an error.

✦ Summary

- ✓ ListView → displays a collection of items.
- ✓ ListViewItem → represents one item.
- ✓ SubItems → adds additional values.
- ✓ ImageIndex → selects the item's image.
- ✓ Items.Add() → adds an item to the ListView.

Thank you 😊